

TEMASEK JUNIOR COLLEGE, SINGAPORE
Year Two
Preliminary Examinations 2008
Higher 2

COMPUTING

9754/02

Paper 2

18 September 2008

Additional Materials: Answer Paper

2.5 hours

READ THESE INSTRUCTIONS FIRST

Write your *Civics Group* and *Name* on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs, music or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **6** printed pages



TEMASEK JUNIOR COLLEGE, SINGAPORE
PASSION PURPOSE DRIVE



Answer **all** questions

Kaguya (家具屋) is a large three storey furniture mall situated in Tampines. The furniture mall is divided into many sections, namely from Section A to Section E. It employs close to 3000 staff members to facilitate the daily smooth running of the mall.

1 Staff members' details are tracked via a database management system.

(a) Explain why a database should be used over a flat file in this case. [3]

(b) Suppose that there are four base relations in the database and they are Manager, Workers, Section, and Shift. Each manager looks after a team of 20 workers each. And each worker can only work in one section. However depending on the shift, the manager can be looking after a different section. A manager is assigned at most two sections to look after. Each section, however, can have only one manager.

(i) Using the above information, draw an E-R diagram to show the relationship between the four tables. [4]

(ii) Explain why an E-R diagram is necessary in database design. [3]

(c) Fields in the Workers table include:

- WorkersID
- Salary

State a suitable data type for these two fields. Justify each of your choices. [4]

(d) The system at Kaguya is able to handle both interactive and batch processing.

(i) Describe the main characteristics of an interactive system. [2]

(ii) The payroll slips of each employee is to be generated at the end of each month. Explain why this application is suitable for batch processing. [3]

- 2 Each furniture item in the store is identified using a code. This code is generated using the following rules.

- o A letter that identifies the section from which the furniture is from. The letters range from A to E.
- o A two digit number ranging from 20 to 99 that identifies the colour.
- o An unlimited string that denotes the country of origin.

For example, E31-SIN represents a piece of furniture of a certain colour from Singapore.

- (a) (i) Using Backus-Naur Form (BNF), define <furniture code>. [5]
- (ii) Hence, draw the syntax diagram for <furniture code>. [3]
- (b) (i) Whenever a transaction is made with regards to a particular piece of furniture, the section id from <furniture code>, and the price of the furniture is noted.
- The price of that furniture is added to an array called SECTION_SALE [X], which would store the total sale for each section. For example, SECTION_SALE[0] would store the total for Section A, SECTION_SALE[5] would store the total for Section E and so on.
- Write a module called CAL_SALE, in pseudo-code, to show how this can be implemented. [2]
- (ii) Write another module called TOTAL_SALE, also in pseudo-code, which makes use the values from SECTION_SALE[X]. This module would calculate and print the total sales figure, together with the section with the highest and lowest sale for each day. [4]
- (c) Kaguya has a loyalty card programme. The system helps keep track of expiring membership in the following way.
- If membership has just expired, a reminder letter would be sent out to members, prompting them to renew their membership.
 - However if the membership expiry is overdue by more than a month, further checks are made as follows
 - o If a customer has been a member for more than 3 years, a warning letter is sent out.
 - o If the customer has spent more than \$2000 within the 1 month, the customer's membership is automatically renewed.
 - o Otherwise, the membership is automatically cancelled and a termination letter is sent out.
- Draw a decision table showing all possible outcomes and results. [6]

- 3** Each customer, after having spent \$2000 or more in a single purchase, is offered a chance to join the loyalty card programme with Kaguya, that offers up to 20% discount on any non-sale items. Each loyalty card number is a seven digit code.

(a) To ensure that a loyalty card number is valid when entered, these card numbers are validated using check digits, where the last digit of the seven digit code is a check digit.

- (i)** Explain what *validation* is. [2]
- (ii)** By using the Modulus-11 system, explain how check digits can be used to check the validity of a loyalty card number. [4]
- (iii)** Suggest a solution that can exclude human errors when entering the loyalty card number. Give **two** reasons to justify your choice. [3]

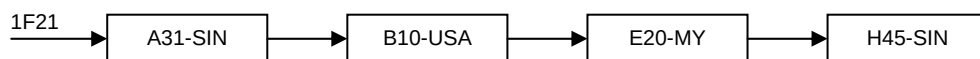
(b) The history of purchases by customers in this mall is kept as a linked-list. Each customer record in the Customer table contains a pointer position to the initial item in the linked-list. A section of the Customer table is shown below

Customer Table

Customer ID	...	Position of Purchase Record
1587439	...	AB10
2131458	...	1F21
1134728	...	235A

Each item in the linked list is sorted in ascending order according to the furniture code described in question 2.

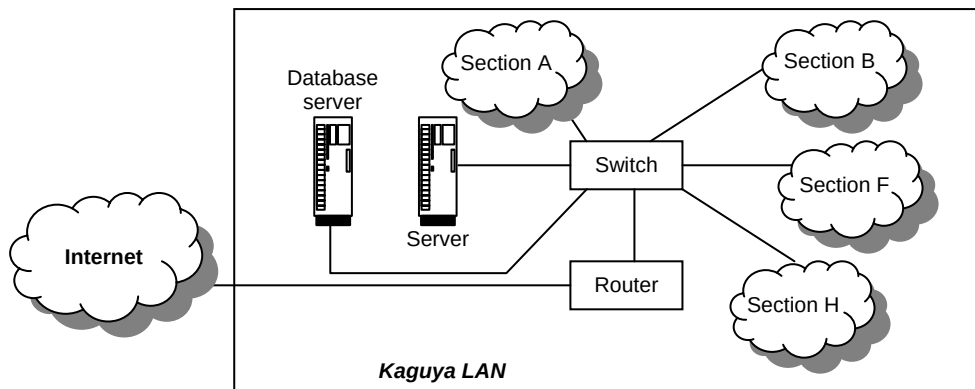
- (i)** Explain how a binary search can be used to locate the correct customer id in the Customer table. [4]
- (ii)** Consider the linked-list below



Explain how a new purchase, coded C21-CAN, can be added to the correct place in the linked-list. [3]

- (iii)** State **two** advantages of using a linked-list system. [2]

- 4 The in-house system is currently linked up via a Local Area Network (LAN) as shown below.



- (a) (i) Describe **three** advantages of having a LAN in Kaguya. [3]
- (ii) Identify the topology used for the LAN at Kaguya. Justify your choice. [3]
- (b) (i) Explain why a router is needed to connect the LAN to the Internet. [3]
- (ii) Describe **two** reasons why a connection to the internet is necessary for the furniture mall. [4]
- (c) Describe **two** security concerns that the LAN administrator at Kaguya might be concerned of, and suggest what actions he/she might take to prevent them. [4]
- 5 A new module to be added to the in-house system was proposed during a routine meeting. This new module to the in-house system would allow customers to access an on-line catalogue from any terminals located within the furniture store. It would also allow customers to query stock availability of existing furniture pieces, on top of prices and specifications.
- (a) Explain the advantages of having an *in-house* system. [3]
- (b) State **two** input devices that should be used in this system. Justify each of your choices. [4]
- (c) Before launching into the analysis phase, a feasibility study is to be conducted.
- (i) Explain the rationale of conducting a feasibility study for the new module. [3]
- (ii) State **two** fact finding methods suitable for this scenario. Justify each of your choices. [4]

- 6** **(a)** Object-Oriented programming has been decided as the paradigm of choice in this project. Explain why object-oriented programming may be a good choice. [3]
- (b)** During testing, it was found that the new module was not able to integrate properly with the existing system. Give **two** reasons why this happened and describe what solutions can be taken. [4]
- (c)** A phased installation will be used to introduce the module into the existing system.
- (i)** Explain what is meant by phased installation. [2]
- (ii)** Would this choice of installation method be suitable? Explain. [3]